Inference Engineering

Step-by-step guide to becoming an Inference Engineer in 2026

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Related Roadmaps

- AI Engineer Roadmap
- ML Roadmap
- C++ Roadmap
- Docker Roadmap
- Kubernetes Roadmap

Learn more through this book

Inference Engineering

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

Inference Engineer

Introduction

- Open vs Closed Models
- Inference vs. Training
- Shared vs. Dedicated Inf.
- The Inference Stack

Basic Terminology

- Time to First Token
- Tokens Per Second
- Inter-Token Latency
- Latency vs. Throughput
- Latency Percentiles
- End-to-End Metrics
- Real-Time Factor

Latency Metrics

- Prefill & Decode Phases
- Online vs. Offline Inf.
- Asynchronous Inference

Inference Phases

- AI-Native Applications

Models & Performance

Neural Networks

- Encoder-Decoder Arch.

Machine Learning

Large Language Models

- Transformer Blocks
- Attention Mechanism
- KV Cache
- Mixture of Experts (MoE)

AI Engineer

Image & Video Models

- Diffusion Models
- Image Generation Arch.
- Few-Step Image Generation
- Video Generation Models

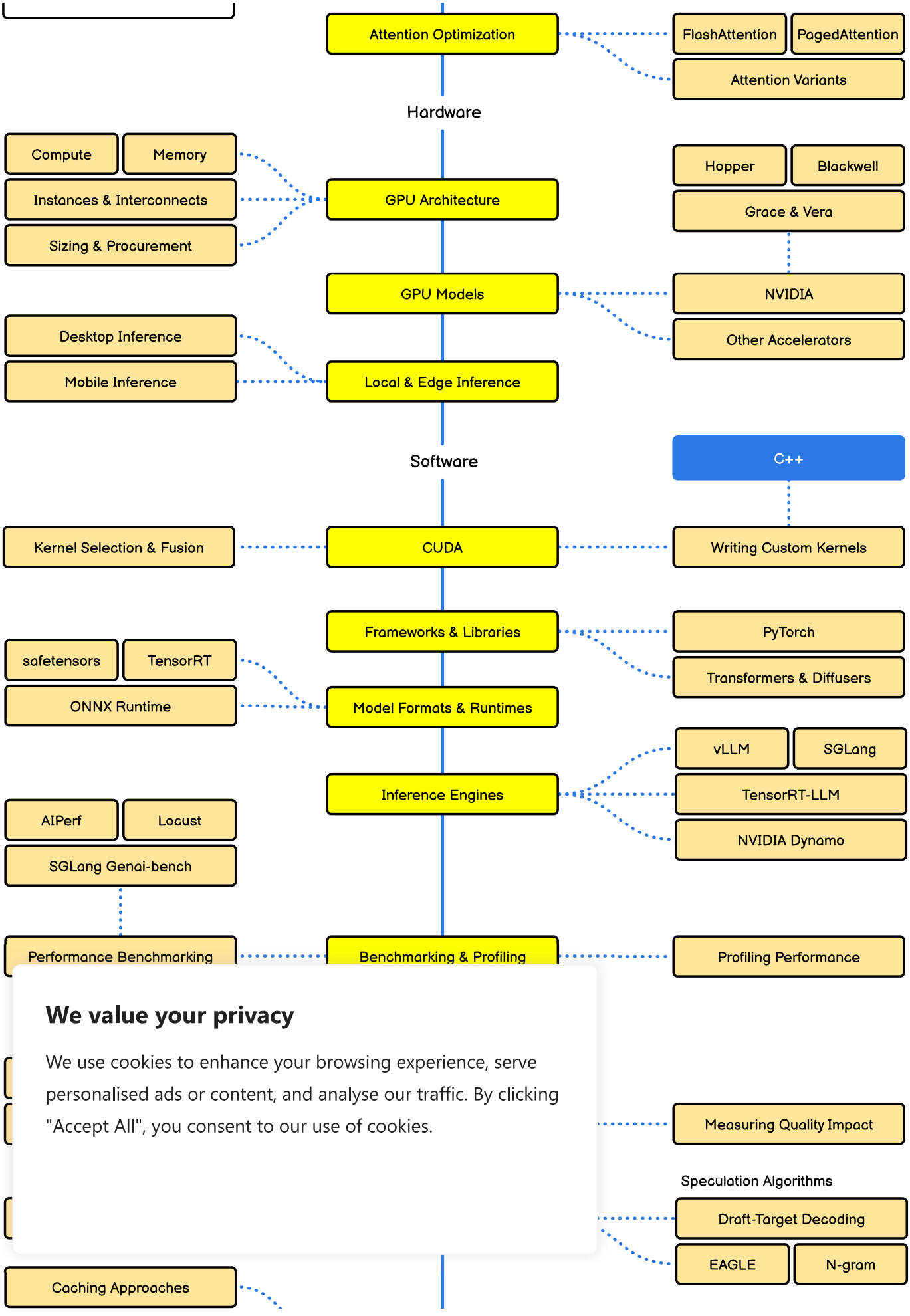
- Ops:Byte Ratio
- Arithmetic Intensity
- Roofline Model
- LLM Bottlenecks
- Image & Video Bottlenecks

Distillation

Model Strategy

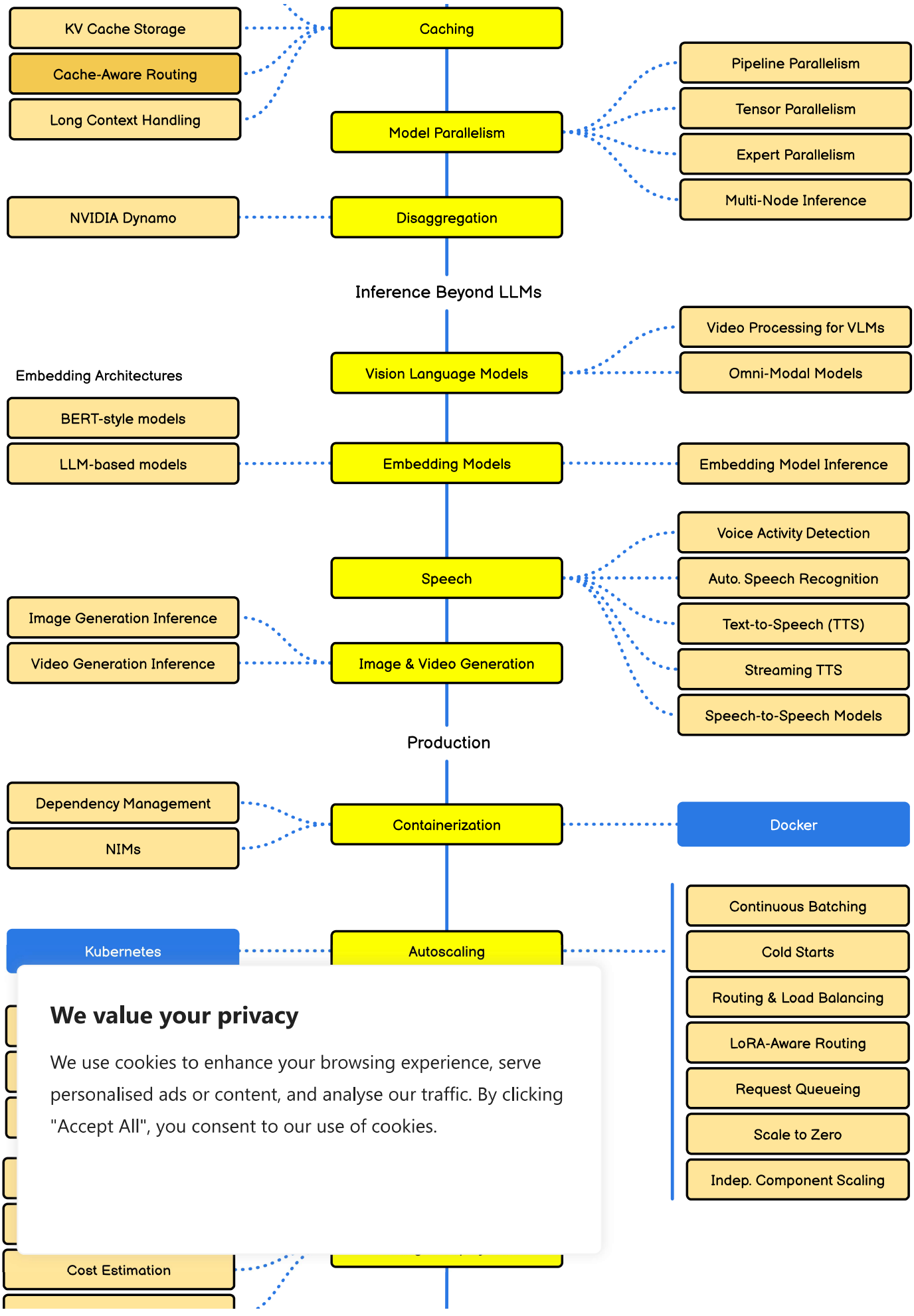
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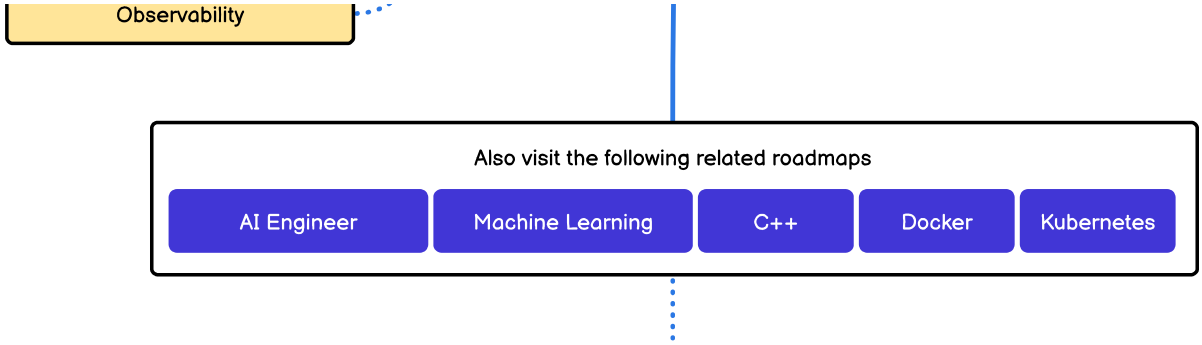
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